

```
PS C:\Users\Naman\Desktop\customer-support-automation> python demo.py
```

```
#####
AI-POWERED CUSTOMER SUPPORT AUTOMATION SYSTEM (LangGraph)
LLM mode: OFFLINE (deterministic, no API key needed)
Customer: CUST-DAVID
#####
```

```
=====
QUERY 1: What are the pricing plans available for your software?
```

```
Expected path: Sales
```

```
-----
Intent classified : Sales
Department       : Sales
Risk category    : none
Requires approval : False
Approval status  : not_required
RAG sources      : pricing_guide.md, pricing_guide.md, pricing_guide.md
Workflow trace:
-> memory: loaded 0 prior interaction(s)
-> intent: Sales | risk: none | engine: rule-based
-> agent: Sales | rag_hits: 3 | sources: pricing_guide.md, pricing_guide.md, pricing_guide.md
-> hitl: low-risk -> auto-approved (no human needed)
-> supervisor: validated & finalized response (approval=not_required)
-> persist: interaction saved to memory.db
-----
```

```
FINAL RESPONSE TO CUSTOMER:
```

```
| Hello,
|
| [Sales] Thanks for reaching out. Based on our records:
|
| Available Plans
|
| If you need anything else regarding this, the Sales team is happy to help.
|
| Best regards,
| ABC Technologies Support Team
```

```
=====
QUERY 2: I forgot my account password.
```

```
Expected path: Account
```

```
-----
Intent classified : Account
Department       : Account
Risk category    : none
Requires approval : False
Approval status  : not_required
RAG sources      : technical_manual.md, faq.md, company_policy.md
Workflow trace:
-> memory: loaded 1 prior interaction(s)
-> intent: Account | risk: none | engine: rule-based
-> agent: Account | rag_hits: 3 | sources: technical_manual.md, faq.md, company_policy.md
-> hitl: low-risk -> auto-approved (no human needed)
-> supervisor: validated & finalized response (approval=not_required)
-> persist: interaction saved to memory.db
-----
```

```
FINAL RESPONSE TO CUSTOMER:
```

```
| Hello,
|
| [Account] Thanks for reaching out. Based on our records:
|
| 2. Login Problems
| - **Forgot password:** click "Forgot password" on the login screen to receive a reset link by email.
| - **Account locked:** after 5 failed attempts the account locks for 15 minutes.
| - **Two-factor (2FA) issues:** ensure your device clock is synced; use a backup code if you lost your device.
|
| If you need anything else regarding this, the Account team is happy to help.
|
| Best regards,
```

=====

QUERY 3: My application crashes whenever I upload a file.

Expected path: Technical Support

Intent classified : Technical
Department : Technical Support
Risk category : none
Requires approval : False
Approval status : not_required
RAG sources : technical_manual.md, technical_manual.md, faq.md
Workflow trace:
-> memory: loaded 2 prior interaction(s)
-> intent: Technical | risk: none | engine: rule-based
-> agent: Technical Support | rag_hits: 3 | sources: technical_manual.md, technical_manual.md, faq.md
-> hitl: low-risk -> auto-approved (no human needed)
-> supervisor: validated & finalized response (approval=not_required)
-> persist: interaction saved to memory.db

FINAL RESPONSE TO CUSTOMER:

| Hello,
|
| [Technical Support] Thanks for reaching out. Based on our records:
|
| Application crashes when uploading a file
| - **Cause:** uploading files larger than the 50 MB limit, or an unsupported file type, can crash older a
pp versions.
| - **Fix:**
| 1. Update to the latest app version (Help → Check for Updates).
| 2. Ensure the file is under 50 MB and of a supported type (PDF, PNG, JPG, DOCX, XLSX, CSV).
| 3. Clear the local cache: Settings → Advanced → Clear Cache.
| 4. If the crash persists, collect logs from Help → Export Diagnostics and contact Technical Support.
|
| If you need anything else regarding this, the Technical Support team is happy to help.
|
| Best regards,
| ABC Technologies Support Team

=====

QUERY 4: I need a refund for my annual subscription.

Expected path: Billing (human approval)

Intent classified : Billing
Department : Billing
Risk category : refund
Requires approval : True
Approval status : approved
RAG sources : company_policy.md, company_policy.md, pricing_guide.md
Workflow trace:
-> memory: loaded 3 prior interaction(s)
-> intent: Billing | risk: refund | engine: rule-based
-> agent: Billing | rag_hits: 3 | sources: company_policy.md, company_policy.md, pricing_guide.md
-> hitl: HIGH-RISK (Refund request) -> requires human supervisor approval
-> hitl: supervisor decision = APPROVED by supervisor(auto)
-> supervisor: validated & finalized response (approval=approved)
-> persist: interaction saved to memory.db

FINAL RESPONSE TO CUSTOMER:

| Hello,
|
| [Billing] Thanks for reaching out. Based on our records:
|
| 1. Refund Policy
| - Customers may request a refund within **30 days** of the original purchase date.
| - Annual subscriptions are eligible for a **prorated refund** for the unused period.
| - Monthly subscriptions are non-refundable once the billing cycle has started, except in cases of proven
service outage.
| - **All refund requests must be reviewed and approved by a human supervisor** before any amount is retur
ned.

```
| - Approved refunds are processed back to the original payment method within **5-7 business days**.  
|  
| If you need anything else regarding this, the Billing team is happy to help.  
|  
| (Note: As this is a sensitive request, it was reviewed and approved by a support supervisor before sending.)  
|  
| Best regards,  
| ABC Technologies Support Team
```

```
=====
```

QUERY 5: What was my previous support issue?

Expected path: Memory recall (no routing)

```
-----
```

Intent classified : Memory
Department : Memory Recall
Risk category : none
Requires approval : False
Approval status : not_required
RAG sources : (none)
Workflow trace:
-> memory: loaded 4 prior interaction(s)
-> intent: Memory (memory-recall query)
-> memory_recall: found 4 record(s)
-> supervisor: validated & finalized response (approval=not_required)
-> persist: interaction saved to memory.db

```
-----
```

FINAL RESPONSE TO CUSTOMER:

```
| Hello,  
|  
| your previous support issue was a Billing request: "I need a refund for my annual subscription." (logged on 2026-06-25T22:24:21).  
|  
| Best regards,  
| ABC Technologies Support Team
```

```
#####
```

Demonstration complete. All interactions stored in:
C:\Users\Naman\OneDrive\Desktop\ibm pro\customer-support-automation\memory.db

```
#####
```

```
PS C:\Users\Naman\Desktop\customer-support-automation> sqlite3 memory.db "SELECT id, customer_id, intent, department, query FROM interactions;"  
id|customer_id|intent|department|query  
1|CUST-DAVID|Sales|Sales|What are the pricing plans available for your software?  
2|CUST-DAVID|Account|Account|I forgot my account password.  
3|CUST-DAVID|Technical|Technical Support|My application crashes whenever I upload a file.  
4|CUST-DAVID|Billing|Billing|I need a refund for my annual subscription.  
5|CUST-DAVID|Memory|Memory Recall|What was my previous support issue?  
PS C:\Users\Naman\Desktop\customer-support-automation>
```